

Mark schemes

Q1.

- (a) Ball is a particle

One appropriate assumption

B1

No air resistance

Second appropriate assumption

B1

2

- (b) (i)

$$0 = 12 \sin 40^\circ - 9.8t$$

$$t = \frac{12 \sin 40^\circ}{9.8} = 0.787 \text{ s}$$

Equation to find time at maximum height

M1

Correct equation

A1

Solving for t

M1

Correct time

A1

4

(ii) $h = 12 \sin 40^\circ \times 0.7871 - 4.9 \times 0.7871^2$
 $= 3.04 \text{ m}$

Substituting time from previous into expression for height

M1

Correct expression

A1

AG; correct height from correct working

A1

3

(c) $2.44 = 12 \sin 40^\circ t - 4.9t^2$

$$4.9t^2 - 12 \sin 40^\circ t + 2.44 = 0$$

$$t = 0.4385 \text{ or } 1.136$$

$$s = 12 \cos 40^\circ \times 1.136 = 10.4 \text{ m}$$

*Equation for time to get to the bar,
based on height being 2.44*

M1

Correct LHS

A1

Correct RHS

A1

Solving quadratic

m1

Correct time / times

A1

*Substituting their larger time into an
expression for the horizontal displacement*

M1

Correct distance

A1

7

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Q2.

(a) $2 = \frac{1}{2} \times 9.8t^2$

*M1: Vertical equation with two terms
including a 2*

A1: Correct equation

M1A1

AG

$$t = \sqrt{\frac{4}{9.8}} = 0.639 \text{ s}$$

dM1: Solving for t

A1: Correct t from correct working

dM1A1

4

(b) $s = 30 \sqrt{\frac{4}{9.8}} = 19.2 \text{ m}$

M1: Use of 30 × time

A1: Correct distance

M1A1

2

- (c) (i) Horizontal remains constant
B1: Correct description

B1

- (ii) Vertical increases in magnitude
B1: Correct description

B1

2

[8]

Q3.

- (a) Ball is a particle/no spin
B1: One assumption

B1

No air resistance/Only gravity or weight
B1: Second assumption

B1

2

- (b) (i) $24.5t - 4.9t^2 = 0$
M1: Equation for vertical motion with height zero

M1

$$\left(t = 0 \text{ or } t = \frac{24.5}{4.9} = 5s \right)$$

AG

A1: Correct equation

A1

dM1: Solving for t

dM1

A1: Correct time from correct working

A1

4

- (b) (ii) $R = 10 \times 5 = 50 \text{ m}$
M1: Use of horizontal component of velocity to find the range

		M1	
	<i>A1: Correct range</i>		
		A1	2
(c)	$20 = 10 t$		
	<i>M1: Horizontal equation</i>		
		M1	
	$t = 2$		
	<i>A1: Time to reach wall</i>		
		A1	
	$h = 24.5 \times 2 - 4.9 \times 2^2 = 29.4 \text{ m}$		
	<i>dM1: Vertical equation for height with $u = 24.5$ and a negative acceleration</i>		
		dM1	
	<i>A1: Correct height</i>		
		A1	4
(d)	No change as acceleration and initial velocity do not change with the mass		
	<i>B1: No change</i>		
		B1	
	<i>B1: Explanation</i>		
		B1	2

Q4.

(a)	$x = 40t \cos \alpha$		
		M1	
	$y = 40t \sin \alpha - \frac{1}{2} g t^2$		
		M1A1	
	$t = \frac{x}{40 \cos \alpha}$		

$$y = 40 \left(\frac{x}{40 \cos \alpha} \right) \sin \alpha - \frac{1}{2} g \left(\frac{x}{40 \cos \alpha} \right)^2$$

m1

$$y = x \tan \alpha - \frac{gx^2}{3200 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{gx^2}{3200 (1 + \tan^2 \alpha)}$$

Answer given

A1

5

$$(b) \quad 4 = 100 \tan \alpha - \frac{9.8 \times 100^2}{3200 (1 + \tan^2 \alpha)}$$

M1

$$245 \tan^2 \alpha - 800 \tan \alpha + 277 = 0$$

M1A1

$$\tan \alpha = \frac{400 \pm \sqrt{(-400)^2 - (245)(277)}}{245}$$

Or equivalent

A1F

M1

$$\alpha = 71^\circ, 21^\circ (22^\circ \text{ acceptable})$$

$$(\text{or } 1.24 \text{ rad.}, 0.375 \text{ rad.})$$

A.T.W.R.T

(or equivalent in radians)

Must be positive

A1F

6

(c) The ball is a particle

E1

No air resistance, etc

E1

2

[13]

Q5.

(a) $0 = (14 \cos 60)^2 - 2 \times 9.8 \times h$

Full method and component of u attempted, v may be present

M1

-1 each error including $v \neq 0$

A2

$h = 2.5 \text{ m}$

AWRT $ft \cos 30$, or $+g$ used

A1F

4

(b) $2.5 = \frac{1}{2} \times 9.8 \times t^2$

*M1: full method for time or half time of flight
ft h , need signs consistent for A1*

M1A1F

$t = 0.714 \text{ sec} \left(\text{or } \frac{5}{7} \right)$

A1F

3

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Q6.

(a) (i) $x = 14t$

*Accept as components of position vector
Accept $x = 15.7 \cos 26.6$*

B1

$y = 7t - 4.9t^2$

Use of equation

M1

3

Substitution

A1

Accept g ; accept $y = 15.7 \sin 26.6t - \frac{1}{2}gt^2$

(ii) $t = \frac{x}{14} \quad y = 7 \times \frac{x}{14} - 4.9 \times \left(\frac{x}{14}\right)^2$

Use of candidate's coordinates

M1

Substituted; accept unsimplified

A1

$$= \frac{x}{2} - 4.9 \frac{x^2}{196}$$

$$y = \frac{20x - x^2}{40}$$

AG; convincingly found

A1

3

(iii) $y = 0 \quad 20x - x^2 = 0$

*Full method for range ($t = 1.4286$ for range,
or $t = 0.7143$ for greatest height and horizontal*

distance then doubled, or $R = \frac{V^2 \sin 2\alpha}{g}$)

M1

$x = 20$

AWRT

A1

2

(b) $x = 12, y = 2.4$

Substitution into appropriate equation(s), ($t = 0.857$)

M1

For $y = 2.4$

A1

$\therefore$ under bar

ft provided $y > 0$

Alt: M1 for $x = 10$ subs and followed by comparison,
 $H = 2.5$ A1, conclusion A1

A1F

3

(c) Air resistance ignored, ball treated as particle...

B1

1

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Q7.

(a) $x = 7t$

B1

$$y = 0 + \frac{1}{2}gt^2$$

Accept ±

M1

$$y = 4.9t^2$$

Accept ±

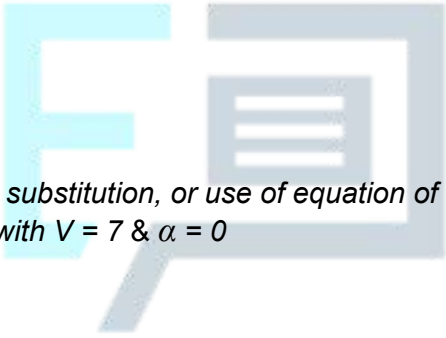
A1

3

(b) $t = \frac{x}{7}$

Attempt at substitution, or use of equation of trajectory with $V = 7$ & $\alpha = 0$

M1


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$$y = \frac{x^2}{10}$$

CAO

A1

2

(c) $8.1 = \frac{x^2}{10}$

Full method for x, accept ±

M1

$$x = 9\text{m}$$

AWRT 9.0 if two stages used

A1

2

(d) vert: $v^2 = 0 + 2 \times 9.8 \times 8.1$

*For M1 Accept $\pm$,
for A1 consistency of signs needed*

M1A1

$u = 7$

B1

speed² = $(7^2 + 12.6^2)$

M1

speed = 14.4 ms^{-1}
(14.414)

A1F

5

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Q8.

(a) (i) vertical: $s = ut + \frac{1}{2} at^2$
 $0 = 7t - 4.9t^2$
Full method

M1

$0 = t(7 - 4.9t)$
 $0 = 7 - 4.9t$

$t = \frac{10}{7} \text{ sec (1.43)}$

A1

3

(ii) $OF = 21 \times \frac{10}{7}$

M1

= 30 m

A1F

2

(b) vert : $0 = 3.5t - 4.9t^2$

Full method required

M1

$$t = \frac{3.5}{4.9}$$

m1

$$FG = 21 \times \frac{3.5}{4.9}$$

M1

$$= 15 \text{ m}$$

A1

4

(c) $GH = \frac{1}{2} \times 15 = 7.5$

B1F

$$OH = 30 + 15 + 7.5$$

M1

$$= 52.5 \text{ m}$$

A1F

3

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[12]

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Q9.

(a) $x = 7t, y = 7t - \frac{1}{2} \times 9.8t^2$ $\left(\begin{matrix} 7t \\ 7t - 4.9t^2 \end{matrix} \right)$

B1 horiz

M1 use of $s = ut + \frac{1}{2}at^2$

M1A1 for vert, accept 10 or g

B1M1A1

3

(b) $t = \frac{x}{7}$

B1

$$y = 7 \times \frac{x}{7} - \frac{1}{2} \times 9.8 \times \frac{x^2}{49}$$

M1A1

$$y = x - 0.1x^2$$

A1

4

Alternative (full verification method)

$x = 7t$ used B1

subs into $y = x - 0.1x^2$

M1A1

$$y = 7t - 4.9t^2 \text{ as in (a)}$$

A1

Alternative: (use equation of path)

$$y = x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$$

45° used

subs

B1

simplify

M1A1

A1

(c) $x = 4, \quad y = 4 - 0.1 \times 4^2$

$$\text{or } t = \frac{4}{7} (0.571(43))$$

M1

$$y = 2.4$$

A1

above wire as $2.4 > 2$

ft slip in y provided y positive

A1F

3

(d) $y = 0$
 $x - 0.1x^2 = 0$

M1

$$x(1 - 0.1x) = 0$$

m1 A1

$$x = 10$$

A1F

4

Alternative: (use of coordinate equations)

$$y = 0, 7t - 4.9t^2 = 0$$

M1

$$\Rightarrow t = \frac{10}{7} \quad (1.43)$$

m1A1

$$x = 7t = 10$$

A1F

(Accept 10.01 if $t = 1.43$ used)

Alternative: (use of symmetry)

$$v = 0, 0 = 7 - 9.8t$$

M1

$$t = \frac{5}{7}$$

A1

$$x = 7 \times t \times 2$$

m1

$$x = 10 \text{ (acc 9.996)}$$

A1F

Alternative: (use of equation for Range)

$$R = \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

Correct formula used M1, 45° used m1

Correct subs A1,

result A1F, AWRT 10m, ft one slip.

(e) $\text{Speed}^2 = 7^2 + 7^2$

M1

$$= 9.90 \text{ ms}^{-1}$$

Accept 9.9, $7\sqrt{2}$,

or 9.91 if $t = 1.43$ used in downward motion

A1

2

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Q10.

(a) $0 = 21 \times 0.7 - 9.8t$

Write 'wrong g ' at first use of 100 or 9.81 and award A0.
complete method with attempt at
component; accept $\alpha = 44.4(3)$; accept $\pm g$

M1 A1

$t = 1.5$

If $\alpha = 44.4$ used, or 2 stage method used,
accept values of t which round to 1.5;
($g = 10$ gives 1.47)

A1

3

(b) (i) $S = 21 \times 0.7 \times 1.5 - \frac{1}{2} \times 9.8 \times 1.5^2$
complete method with attempt at
component; accept c 's time and $\pm g$

M1

or $0 = (21 \times 0.7)^2 - 2 \times 9.8 \times S$
all correct, (accept c 's time) including $-g$ and $\sin \alpha$

A1

$S = 11.025$

Accept c 's time; ft one slip e.g. $+g$ or $\cos \alpha$

A1F

Distance above tree = $11.025 - 8 \approx 3\text{m}$

ft if $d > 0$,

$g = 10$ gives $S = 10.8$ and $d = 2.8$ (accept)

A1F

4

(ii) $S = 8$

$$8 = 21 \times 0.7t - \frac{1}{2} \times 9.8 \times t^2$$

full method for t or v with attempt at component ;
accept $\pm g$

all correct, including $-g$ and $\sin \alpha$,
but condone repeated error in $-g$ and $\sin \alpha$ from (i)

M1 A1

$$4.9t^2 - 14.7t + 8 = 0$$

$$t = \frac{14.7 \pm \sqrt{14.7^2 - 4 \times 4.9 \times 8}}{2 \times 4.9}$$

To find t by solving equation or using $v(=7.7)$

in $v = u + at$

Substituting; ft

m1 A1F

$$t = 0.71 \text{ (0.7143)}$$

CAO

A1

5

(iii) time = $2.29 - 0.71$

Subtraction of two positive values of t ,
or use of symmetry

M1

or $2(1.5 - 0.71)$

Allow values between 2.28, 2.29 and 0.71, 0.72

$$= 1.57 \text{ sec}$$

Accept 1.56 – 1.58 ($g = 10$ gives $t = 1.5$)

A1F

2

Alternative: If $\cos 44.4$ used:

(b)(i) $S = 11.475$, $d = 3.475$

(ii) $t = 0.688$ (2.373)

(iii) $t = 1.69$

If no component:

(a) M0;

(b)(i) SC: $d = 12.5$

M1 A1 for complete method

(ii) SC: $t = 0.423$

M1 A1 for complete method

(iii) $t = 3.42$

M1A1

[14]

Q11.

(a) Vertical: $0 = 4Ut - \frac{1}{2}gt^2$
Accept use of 9.8

M1A1

$$t = \frac{8U}{g}$$

A1

3

(b) horizontal: $OH = 3U \times \frac{8U}{g}$
Use of $v \times t$

M1

$$OH = \frac{24U^2}{g}$$

A1

2

(c) $\text{speed}^2 = (3U)^2 + (4U)^2$

M1

$$\text{speed} = 5U$$

A1

2

(d) $(\sqrt{18}U)^2 = (3U)^2 + v^2$

M1A1

$$v = (\pm)3U$$

A1

$$\pm 3U = 4U - gt$$

Or attempt to solve quadratic in t

M1

$$t_1 = \frac{U}{g} \qquad t_2 = \frac{7U}{g}$$

A1A1

6

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Q12.

(a) $25\cos 17^\circ t = 12$

*Equation to find time of flight
Correct equation*

M1 A1

$$t = \frac{12}{25\cos 17^\circ} = 0.502 \text{ s}$$

ag Correct time from correct working

A1

3

(b) $h = 25 \sin 17^\circ \times 0.502 - 4.9 \times 0.502^2$

*Substituting $t = 0.502$ into an expression
for the height of the ball*

M1

Correct expression

A1

$$= 2.43 \text{ m}$$

Correct height

A1

3

(c) $v_x = 25\cos 17^\circ$

*Finding horizontal and vertical
components of velocity*

M1

$$v_y = 25\sin 17^\circ - 9.8 \times 0.502 = 2.390$$

Horizontal component correct

A1

Vertical component correct

A1

$$\tan \alpha = \frac{25 \cos 17^\circ}{2.390}$$

Use of tan to find angle

M1

$$\alpha = 84.3^\circ$$

Correct angle

A1

5

- (d) Ball is a particle

First assumption

B1

No air resistance

Second assumption

B1

2

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Q13.

- (a) $8 = 10 \cos 60^\circ t$

Horizontal equation based on travelling 8 metres

M1

Correct equation

A1

$$t = \frac{8}{10 \cos 60^\circ} = 1.6 \text{ s}$$

ag *Correct time from correct working*

A1

3

- (b) $h = 10 \sin 60^\circ \times 1.6 - 4.9 \times 1.6^2$

Expression for height using $t = 1.6$

M1

Correct expression

A1

Correct height

A1

(c) $V = 10 \cos 60^\circ = 5$

Horizontal component of the velocity

B1

$$V = 10 \sin 60^\circ - 9.8 \times 1.6 = -7.020$$

Expression for vertical component of velocity

M1

Correct vertical component

A1

Finding the magnitude

M1

$$v = \sqrt{5^2 + 7.020^2} = 8.62 \text{ ms}^{-1}$$

Correct magnitude

A1

5

[11]

Q14.

(a) $0 = 10 \sin 70^\circ t - 4.9t^2$

Equation for height equal to zero

Solving for t

M1 M1

$$t = 0 \text{ or } t = \frac{10 \sin 70^\circ}{4.9} = 1.918 \text{ s}$$

Correct t

Calculating range

A1 M1

$$R = 10 \cos 70^\circ \times 1.918 = 6.56 \text{ m}$$

Correct range

A1

5

(b) $-2 = 10 \sin 70^\circ - 4.9t^2$

Forming equation for vertical motion

when ball lands

LHS correct

RHS correct

M1 A1 A1

$$4.9t^2 - 10 \sin 70^\circ t - 2 = 0$$

$$t = 2.11 \text{ or } -0.193$$

Solving quadratic equation

Correct solution

M1A1

$$R = 10 \cos 70^\circ \times 2.11 = 7.22\text{m}$$

Calculating range

Correct range

M1A1

7

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Q15.

(a) $5 = 32\sin 60^\circ t - 4.9t^2$

M1

Equation for vertical motion with ± 5

M1

LHS correct

A1

RHS correct

A1

$$4.9t^2 - 32\sin 60^\circ t + 5 = 0$$

$$t = 0.1866 \text{ or } 5.4691$$

Solving quadratic

m1

5.47 seconds

Selecting larger answer from two solutions or obtaining one answer with a reason

A1

5

(b) $32\cos 60^\circ \times 5.469 = 87.5 \text{ m}$

Equation for horizontal motion

M1

Correct range

A1

2

(c) $v_H = 32\cos 60^\circ$

Horiz. component of velocity seen or used

B1

$$v_V = 32\sin 60^\circ - 9.8 \times 5.469 = -25.88$$

Finding vertical component of velocity

M1

Correct vertical component

A1

$$v = \sqrt{16^2 + 25.88^2} = 30.4 \text{ ms}^{-1}$$

Finding magnitude

m1

Correct speed

(Note Max Height = 39.2 m from $t = 2.83$)

A1

5

[12]

Q16.

(a) (i) $20\sin 30^\circ t - 4.9t^2 = 0$

Forming equation for time of flight

M1

Correct equation

A1

$$t = 0 \text{ or } t = \frac{20\sin 30^\circ}{4.9} = 2.04 \text{ (to 3 sf)}$$

Solving quadratic equation

M1

ag *Correct solution from correct working*

A1

4

(ii) $R = 20 \cos 30^\circ \times 2.04 = 35.3 \text{ m}$

Calculation of range

M1

Correct range

A1

2

(b) $20 \sin 30^\circ t - 4.9t^2 = 2$

Equation to find t at height of 2

M1

Correct equation

A1

$$4.9t^2 - 10t + 2 = 0$$

$$t = 0.2248 \text{ or } t = 1.82$$

Correct times

A1

$$v_y = 20 \sin 30^\circ - 9.8 \times 0.2248 = 7.797$$

Calculating vertical component

M1

Correct vertical component

A1

$$v_x = 20 \cos 30^\circ = 17.32$$

Finding horizontal component

B1

Finding speed from components

M1

$$v = \sqrt{v_x^2 + v_y^2} = 19.0 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Correct speed

A1

8

[14]

Q17.

(a) $0 = V \sin \alpha - gt$

Equation for t at max. height

M1

$$t = \frac{V \sin \alpha}{g}$$

Solving for t

M1

Correct t

A1

$$H = V \sin \alpha \times \frac{V \sin \alpha}{g} - \frac{1}{2} \times g \times \left(\frac{V \sin \alpha}{g} \right)^2$$

Substitution into equation for height

M1

Correct expression

A1

$$= \frac{V^2 \sin^2 \alpha}{2g}$$

Correct final result from correct working

A1

6

(b) (i) $6 = \frac{V^2}{2g} \times \frac{3}{4}$

Substitution into given formula

M1

$$V = \sqrt{16g} = 12.5 \text{ ms}^{-1}$$

Solving for V

M1

Correct value

A1

3

(ii) $t = \frac{2\sqrt{16g} \sin 60^\circ}{g} = 2.213$

Finding time of flight

M1

Correct time of flight

A1

$$R = \sqrt{16g} \cos 60^\circ \times 2.213 = 13.9\text{m}$$

Finding range

M1

Correct range

A1

4

[13]

Q18.

(a) $0 = V \sin \alpha t - gt^2$

Equation for height equal to zero

B1

$$t = 0 \text{ or } t = \frac{2V \sin \alpha}{g}$$

Finding t

Correct time of flight

M1 A1

$$\text{Range} = V \cos \alpha \times \frac{2V \sin \alpha}{g}$$

Expression for horizontal displacement

Substituting the time of flight

M1 M1

$$= \frac{V^2 \sin 2\alpha}{g}$$

Correct final answer from correct working

A1

6

(b) Because the maximum value of $\sin 2\alpha$ is 1.

Reason that uses maximum of sin being 1

B1

$$\alpha = 45^\circ$$

$$45^\circ$$

B1

2

(c) (i) $\frac{V^2}{g} = 35$
Substituting 35

M1

$V = \sqrt{35g}$
cao = 18.5 m s⁻¹

A1

2

(ii) $R = \frac{18.5^2 \sin 60^\circ}{g} = 30 \text{ m}$
Substituting $\alpha = 30^\circ$
30 m

M1 A1

2

(or $R = 35 \sin 60^\circ = 30$)

[12]

Q19.

(a) $-25 = 40 \sin 5^\circ t - 4.9t^2$

M1/A1/A1

$4.9t^2 - 40 \sin 5^\circ t - 25 = 0$

Equation for vertical motion

LHS correct

RHS correct

$t = \frac{40 \sin 5^\circ \pm \sqrt{(40 \sin 5^\circ)^2 - 4 \times 4.9 \times (-25)}}{2 \times 4.9}$

Solving quadratic equation

Correct method

m1/A1

= 2.642 or -1.931

Hits water after 2.64 seconds

Selecting 2.64

A1

6

- (b) $x = 40\cos 5^\circ \times 2.642 = 105 \text{ m}$
Equation for horizontal motion
Correct range

M1/A1

2

- (c) $v_x = 40\cos 5^\circ$

M1/M1/A1

$v_y = 40\sin 5^\circ - 9.8 \times 2.642$
Horizontal component at impact
Vertical component at impact
Vertical component correct

$v = \sqrt{(40 \cos 5^\circ)^2 + (40 \sin 5^\circ - 9.8 \times 2.642)^2}$
Finding magnitude

M1

$= 45.7 \text{ ms}^{-1}$
Correct final answer

A1

5

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